BMP2MSX

BMP TO MSX

IMAGE CONVERTER

BMP2MSX USER'S MANUAL

2026.11

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Chapter 1 Introduction

1.1 What Is BMP2MSX

BMP2MSX (shown on screen as "BMP to MSX") is a tool that converts Windows image files into the BSAVE format, one of the image formats used on the MSX. Its purpose is to let you create images with the rich hardware and software available on Windows, and then load and display them on an MSX.

Because Windows and the MSX differ greatly in display capability, BMP2MSX includes features to absorb the difference in dot aspect ratio between Windows and the MSX, and to reproduce the original colors as faithfully as possible within a limited number of colors, through color reduction and palette selection. Since the output is in BSAVE format, it can be loaded by many MSX graphics tools, which makes fine adjustment after conversion relatively easy.

Even so, the conversion is done by machine and has its limits. If you are aiming for the highest quality, we recommend fine-tuning while checking the colors on a real MSX.

This manual describes BMP2MSX ver 2026_10_13 ((C)2000-2026 HRA!). There are two builds, Japanese (BMP2MSX.exe) and English (BMP2MSXE.exe); they differ only in display language and are functionally identical. The English edition of this manual follows the wording of the English build, and the Japanese edition follows the Japanese build.

Note that the Help menu and the HTML Help file (bmp2msx.chm) found in older versions have been retired; the usage documentation now lives in this manual (PDF). Version information is available from the "Version..." menu item (Section 1.4).

1.2 Features

The main features of BMP2MSX are as follows.

Feature	Description
Image loading	In addition to uncompressed BMP (1/4/8/16/24/32 bpp), JPEG, PNG and other BMP files can be loaded through the Windows Imaging Component (WIC). Clipboard images can also be

Feature	Description
	converted (Chapter 3).
BSAVE output	Outputs BSAVE-format images for SCREEN 2/3/5/6/7/8/10/12. The 256-line modes and interlaced output are also supported (Chapter 6).
BAS output	Can also output an MSX-BASIC program (.BAS) that loads and displays the converted result (Section 6.5).
Palette output	Palette data can be output in Graph Saurus format (PL5/PL7) or CSV format, or embedded in the BSAVE image (Section 6.4).
V9968 output	Can output images with a 256-color palette for the V9968's extended palette mode (EPAL) (Chapter 7).
Palette selection	Part or all of the palette can be decided automatically; manual setting, disabling, and sorting are also possible (Chapters 5 and 7).
Color reduction	Reduction is possible with several methods, including error diffusion and dithering (Section 5.3).
Preview	The converted result can be checked on screen as an MSX display image. Up to 50 previews are retained and can be switched for comparison (Chapter 4).
Original image view	A separate window shows the image before size adjustment, with an eyedropper to inspect color values (Chapter 11).
Scripts	Fixed conversion procedures can be semi-automated with text scripts (Chapter 9).
Size adjustment	Resizing that takes the aspect ratio into account (Section 5.4).
Color calibration	The conversion tables that compensate for the difference in appearance between a real MSX and Windows can be adjusted (Section 8.3).
Command-line version	A command-line version for Linux (bmp2msx-cli) is available, which performs the same conversions as the Windows version, including running scripts (Chapter 10).

1.3 Installation and Files

BMP2MSX has no installer. Just extract the distribution archive and place BMP2MSX.exe (or BMP2MSXE.exe for the English build) in any folder. The latest version is distributed on the bmp2msx repository on GitHub (https://github.com/hra1129/bmp2msx).

The following files are placed next to the executable (some are generated automatically).

File	Contents
BMP2MSX.exe / BMP2MSXE.exe	The executable (Japanese / English build).
BMP2MSX.B2M	The settings file. The default settings are saved automatically on exit and loaded at the next start. The files used by "Save as" and "Load" in the settings dialog are also in this format. The registry is not used.
BMP2MSX_CUSTOM.BIN	Stores the color calibration adjustment data.
Load.txt	A collection of sample programs that display converted results in MSX-BASIC (Chapter 12).

Because the settings are stored next to the executable, placing BMP2MSX in several folders lets each copy keep its own default settings.

1.4 Checking the Version

Choosing "Version..." at the right end of the main window's menu opens the version dialog, which shows the version (for example, ver 2026_10_13 (C)2000-2026 HRA!). Please check that your copy of BMP2MSX matches the version this manual covers.

1.5 How This Manual Is Organized

This manual is organized as follows. Chapter 2 is a tutorial for first-time users. Once you have grasped the overall flow, you can read Chapters 3 onward only where needed.

Chapter	Contents
Chapter 1 Introduction	Overview of BMP2MSX and its files.
Chapter 2 Tutorial	The flow from preparing an image to displaying it on an MSX.
Chapter 3 Loading Image Files	Supported formats and how to load images.
Chapter 4 The Main Window	Layout of the main window, its menus, and preview operations.
Chapter 5 The Settings Dialog	Every conversion setting explained.
Chapter 6 Screen Modes and Output Files	The list of output modes, and the specifications of BSAVE, palette files and BAS output.
Chapter 7 Output for the V9968	Conversion and output for the V9968's EPAL mode.
Chapter 8 Palette Editing and Color Calibration	Manual palette editing and color compensation.
Chapter 9 Scripts	The script window and a reference of script commands.
Chapter 10 The Command-Line Version	Building and using the Linux bmp2msx-cli.
Chapter 11 The Original Image Window	Displaying the original image and the eyedropper.
Chapter 12 Displaying the Result on an MSX	Example programs that display converted results on an MSX.
Chapter 13 Distribution and Support	Copyright, distribution terms, disclaimer and support.

In this manual, screen modes are written as in the MSX-BASIC SCREEN statement, for example SCREEN 2. Options shown as "SCREEN2/4" or "SCREEN10/11" in the settings dialog indicate that those modes share the same image format (Section 6.1).

Chapter 2 Tutorial

BMP2MSX has many features, which can be a little overwhelming at first. This chapter shows the full flow: prepare an image, convert it for SCREEN7, and display it on an MSX. For the detailed meaning of each setting, see Chapter 5.

2.1 Prepare an Image

First, prepare the image file to convert. BMP2MSX can load JPEG and PNG directly, as well as uncompressed BMP (Section 3.1).

Here we will use the following image (640x480).



It is convenient to keep the image at a manageable size such as 640x480. BMP2MSX has a size adjustment feature, but it works by forcibly matching the MSX resolution. So if you plan to use size adjustment, keep the original image at the same aspect ratio it will have when displayed, so that the converted result is not distorted.

2.2 Change the Settings

When BMP2MSX starts, the main window appears. The current settings are shown at the bottom left of the window; in the initial state the screen mode should read "SCREEN5". To create an image for SCREEN7, this setting must be changed.

Choose Setting → Setting... from the menu to open the default settings dialog, and specify the following.

- Select SCREEN7 under "MSX screen mode"
- Set the dither pattern to "net dots", check error diffusion, and set the error coefficient (strength) to 0.430
- Check size adjustment, select "antialiasing" for resampling and "output size forcibly" for the size mode
- Leave everything else as it is

Then click the "Default setting" button to close the dialog. The summary at the bottom left of the main window should now show SCREEN7.

2.3 Convert

Drag the image file's icon from Explorer and drop it onto the main window. Conversion starts, and after a short while the converted MSX display image (the preview) appears.

The preview looks slightly stretched vertically. This is because the dot aspect ratio differs between the MSX and Windows. The preview is drawn with the Windows dot ratio, but MSX dots are slightly wider than they are tall, so when the image is displayed on an MSX, the original proportions are reproduced.

If the output destination is "file and preview" and "Auto file name" is checked, a converted file (with the extension .SC7 in this example) is also created in the same folder as the source file.

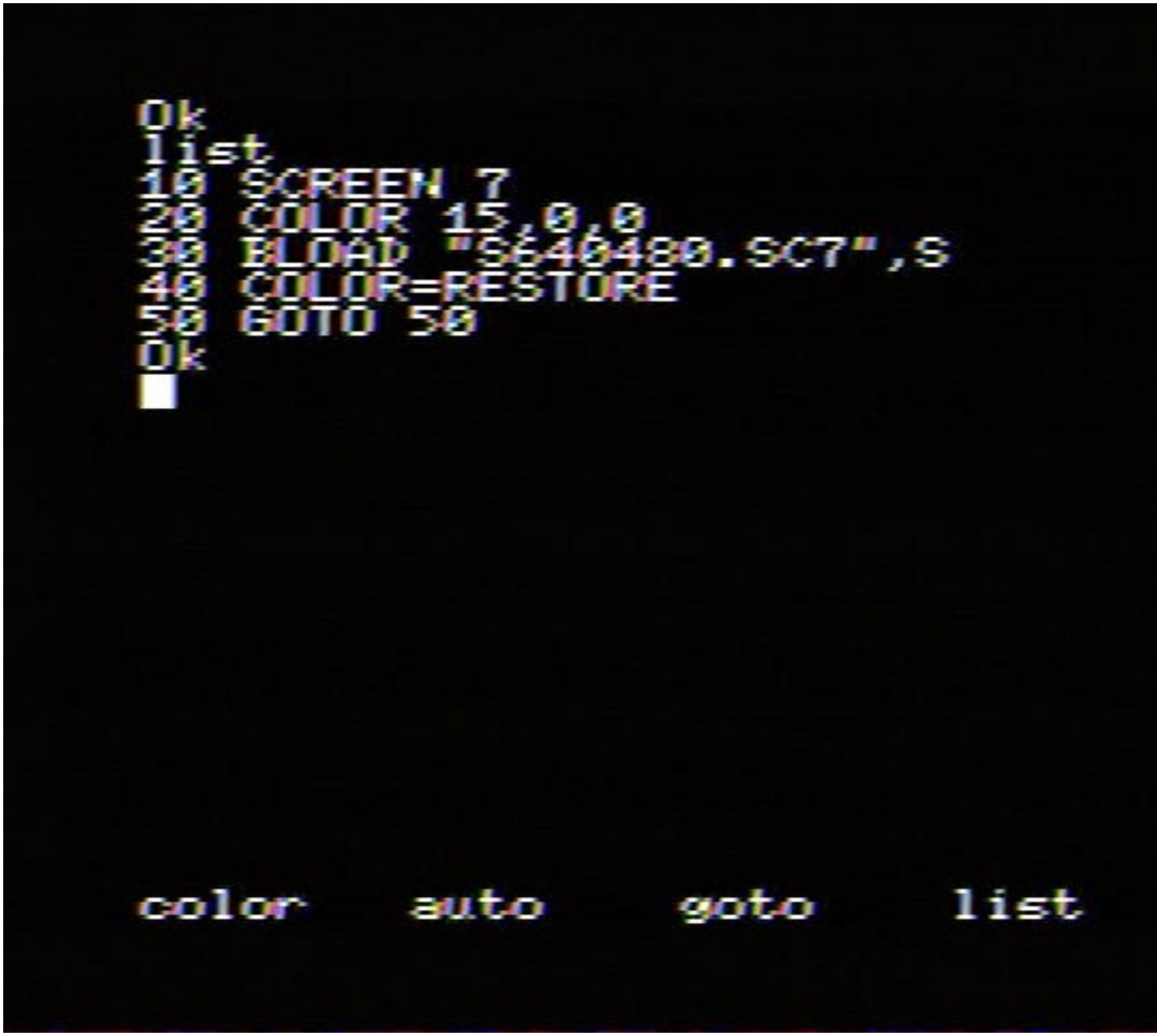
If you do not use automatic file naming, a save dialog appears for each conversion. Dropping several files at once converts them in sequence.

2.4 Display It on an MSX

See Chapter 12 for the details of displaying the result on an MSX. Here we only show the outcome.



Put the converted result on a floppy disk or similar medium, and check that MSX-BASIC can see it.



Enter and run a simple program that loads the image.



The image, which looked stretched vertically in the Windows preview, is displayed on the MSX in its original proportions. Note that the actual colors vary with the capture and display environment; always do the final check on real hardware.

Chapter 3 Loading Image Files

3.1 Supported Formats

BMP2MSX can load the following inputs.

Format	Notes
Windows BMP files (1/4/8/16/24/32 bpp, uncompressed)	Uncompressed 24 bpp BMP files are loaded directly through the traditional path. Other BMP files are loaded through the Windows Imaging Component (WIC).
JPEG files	Loaded through WIC.
PNG files	Loaded through WIC.
Clipboard images	Can be converted from the Edit menu (Section 3.2).

Images loaded through WIC are converted internally to a 24-bit format before processing. Images that WIC cannot handle, such as BMP files in compression formats it does not support, cannot be loaded. In that case, convert them beforehand to uncompressed BMP or PNG with another tool.

3.2 How to Load an Image

There are four ways to load an image. In every case, conversion takes place as soon as the image is loaded.

Method	Procedure
Drag and drop	Drop image files onto the main window. Dropping several files at once converts them in sequence.
Open from the menu	Choose File → Open file and select an image file.

Method	Procedure
Drop onto the icon	Drop image files onto the BMP2MSX icon (the executable) to start it. Conversion uses the default settings.
Clipboard	Choose Edit → Convert from clipboard (Ctrl+V) to convert the image held in the clipboard.

3.3 What Happens After Loading

A loaded image is converted immediately. If the output destination includes a preview, the MSX display image appears in the main window (if the output destination is set to file output only, no preview is created).

Which settings are used for the conversion follows the "Use setting" option of the settings dialog (Section 5.2). The default settings are used for the first conversion.

Before conversion, the image is fitted to the MSX screen size. If size adjustment is enabled, the image is scaled to the MSX resolution. If size adjustment is disabled, an image larger than the MSX resolution is cropped to the area that fits, starting from the top left, and any missing area is filled with the background color. See Section 5.4 for details.

The white text on blue at the bottom left of the main window shows the settings used for that conversion (Section 4.2).

Chapter 4 The Main Window

4.1 Layout

The main window is the central window of BMP2MSX. Right after startup it shows the title logo and the message "Drop BMP files on this window". It consists of the image area and the settings summary at the bottom left.

Files can be dropped onto this window to convert them (Chapter 3). Clicking the image area has the following actions assigned.

- Clicking while the title logo is shown opens the default settings dialog
- Clicking while an MSX display image (preview) is shown opens the reconvert dialog (Section 4.5)

4.2 The Settings Summary

The settings are shown in text at the bottom left of the window. While the title logo is shown, it displays the default settings; while a preview is shown, it displays the settings used to convert that image.

The background color has a meaning.

Background	Meaning
Blue	Showing a conversion result (MSX display image). The title logo state also uses this color.
Red	Showing the source image (switched with the Space key; Section 4.4).

This display can be shown or hidden with Setting → Show setting (Tab key).

4.3 Menus

The main window has the following menus.

Menu	Contents
File	Basic operations such as loading files and exiting.
Edit	Converting an image from the clipboard.
Preview	Operations on the displayed MSX display image.
Setting	Operations on the default settings, and color calibration.
Version...	Opens the version dialog (Section 1.4). This is a single menu item, not a menu with items inside.

File menu

Item	Shortcut	Description
Open file	None	Shows a dialog for opening an image file.
Script mode	None	Shows the script window (Chapter 9).
Exit	None	Exits BMP2MSX.

Edit menu

Item	Shortcut	Description
Convert from clipboard	Ctrl+V	Converts the image held in the clipboard.

Preview menu

Item	Shortcut	Description
View → Next	→	Shows the next preview image.
View → Prev	←	Shows the previous preview image.

Item	Shortcut	Description
View → Change original / result	Space	Switches between the source image and the preview.
View → Show original	Ctrl+O	Shows the complete source image before size adjustment (Chapter 11).
View → Zoom up	F7	Enlarges the display (up to 4x).
View → Zoom down	F8	Reduces the display.
Clear now preview	Del	Removes the displayed preview from memory.
Clear all preview	Ctrl+A	Removes all retained previews.
Save	None	Saves the displayed preview in BSAVE format.
Save as	Ctrl+S	Saves the displayed preview in BSAVE format under a chosen name.
Save BMP	None	Saves the displayed preview in BMP format (Section 6.6).
Reconvert	Ctrl+R	Converts the displayed preview again (opens the reconvert dialog).

Setting menu

Item	Shortcut	Description
Setting...	Ctrl+E	Shows the default settings dialog (Chapter 5).
Show setting	Tab	Shows or hides the settings summary at the bottom left of the main window.
Load default	None	Resets the default settings to their initial values.
Color calibration	None	Shows the color calibration dialog (Section 8.3).

4.4 Working with Previews

A converted MSX display image is called a preview. Up to 50 previews are retained. With the main window focused, the left and right cursor keys (the up and down keys also work) switch the displayed preview. The Preview → View → Next / Prev menu items do the same.

While a preview is shown, pressing the Space key switches between the source image (the full-color image right after size adjustment) and the MSX display image. This is convenient for comparing before and after conversion. While the source image is shown, the background of the settings summary at the bottom left turns red (Section 4.2). Pressing Space again returns to the MSX display image.

The complete original image, before size adjustment, can be viewed with Preview → View → Show original (Ctrl+O); see Chapter 11.

The display magnification can be switched from 1x to 4x with F7 (zoom up) and F8 (zoom down).

4.5 Reconverting

Clicking the image area while an MSX display image is shown, or choosing Preview → Reconvert (Ctrl+R), opens the reconvert dialog. It shows the settings that were used to convert that image; change the settings and click the "Reconvert" button to convert again with the new settings. See Section 5.6 for details of the reconvert dialog.

Chapter 5 The Settings Dialog

5.1 Basics

The settings dialog tells BMP2MSX what kind of conversion to perform. Because Windows and the MSX differ greatly in display capability, good results require detailed settings that suit the image. This dialog is where those settings are made.

There are two forms of the dialog: the default settings dialog and the reconvert dialog (for the difference, see Section 5.6). Both are opened from the main window.

- Default settings dialog: Setting → Setting... (Ctrl+E). It also opens when the image area is clicked while the title logo is shown
- Reconvert dialog: click the image area while a preview is shown, or Preview → Reconvert (Ctrl+R)

There are many settings, but depending on the MSX screen mode, some items are enabled and others are not. Unavailable items are grayed out. For example, in SCREEN8 and SCREEN12, which use no palette, most palette settings are disabled; in SCREEN10 and SCREEN12, which use YJK, most palette-related items other than the natural image algorithm are disabled.

The buttons at the top of the dialog are as follows.

Button	Description
Save as	Save the displayed settings to a file (extension .B2M).
Load	Load a saved B2M file and apply it to the displayed settings.
Default setting (labeled "Reconvert" in the reconvert dialog)	In the default settings dialog, adopts the displayed contents as the default settings and closes the dialog. In the reconvert dialog, converts again with the displayed contents (Section 5.6).
Cancel	Discards all changes and closes the dialog.
Copy to default	Shown only in the reconvert dialog. Copies the displayed reconvert settings to

Button	Description
	the default settings.

5.2 Output Mode Settings

These are specified in the "Output mode" group at the top of the dialog.

MSX screen mode

Specifies which MSX screen mode to output for. The choices follow the MSX-BASIC SCREEN statement: SCREEN2/4, SCREEN3, SCREEN5, SCREEN6, SCREEN7, SCREEN8, SCREEN10/11, SCREEN12, and the 256-line version of each of SCREEN5 through SCREEN12. See Section 6.1 for the specification of each mode.

Output destination

Specifies where the conversion result goes.

Option	Description
File	Creates the BSAVE-format file of the conversion result. No MSX display image (preview) is created.
Preview	Creates only the MSX display image; no file is output. Use this for trial conversions.
File and preview	Creates both the MSX display image and the BSAVE-format file.

192-line mode

When checked, SCREEN5 and above are processed as 192 vertical lines (384 lines when interlaced). This generates images for use with the setting VDP(10) = VDP(10) AND &B01111111.

Interlace

Specifies whether to use the MSX interlace mode. When checked, two image files are generated for interlace mode (Section 6.3). Loading them and putting the MSX into interlace mode gives a pseudo

doubling of the vertical resolution.

Auto file name

When checked, and the output destination creates files, file names are assigned automatically and you are no longer asked for a save file name. See Section 6.3 for the naming rules. Note that an existing file of the same name is overwritten.

BAS output

When checked, an MSX-BASIC program (.BAS) that loads and displays the conversion result is output at the same time (Section 6.5).

Use setting

Selects which settings are adopted when converting a new image file.

Option	Description
Default	Every conversion, including the second and later ones, uses the default settings.
Last convert	The first conversion uses the default settings; the second and later ones use the settings of the last conversion.
Indication image	The first conversion uses the default settings; the second and later ones use the settings of the displayed MSX display image if there is one, and the default settings otherwise.

Always copy reconvert to default

A check box in the reconvert dialog. When checked, the settings of each reconversion are copied to the default settings at that moment.

5.3 Conversion Settings

The "Conversion settings" group in the middle of the dialog mainly determines image quality.

Dither pattern

Dithering arranges similar colors in a mesh pattern to approximate intermediate colors and seemingly increase the number of colors. This option selects the shape of the mesh pattern. The choices are as they appear in the English build's list.

Option	Description
no use	Does not apply a dither pattern.
random dots	Uses a random-looking pattern.
net dots	Uses an organized mesh pattern.
net dots (double length)	An organized mesh pattern stretched vertically. Combined well, it can suppress flicker in interlace mode.
pattern of brick	A pattern like a brick wall. It has few uses.
pattern of grain	A somewhat organized mottled pattern.
rough net	Uses an organized mesh pattern, but luminance changes become strong.
lite net	Uses an organized mesh pattern, but luminance changes are weak. In modes such as SCREEN10 the mesh is less noticeable.
half tone	A mesh of an exact 50% ratio.

Dither adding method

The dither pattern is applied to each of the R, G and B components. This option decides how the pattern added to each component is shifted.

Option	Description
rotate	Each of R, G and B uses the pattern rotated by 90 degrees from the next. The mesh is not very noticeable.
pile up	Exactly the same mesh is applied to R, G and B. The mesh is very noticeable.
moveX	Shifts the pattern one step right for each of R, G and B. When using vertical net dots to suppress interlace flicker, this method must be selected or the effect is lost.
moveY	Shifts the pattern one step down for each of R, G and B.

In general, "rotate" seems to give good results.

Color reproduction accuracy

The loaded image is treated as full color, and just before color reduction starts, the number of bits of each of R, G and B is reduced in advance. The choices are 8bit (no reduction), 7bit (ignores the lowest 1 bit), 6bit (lowest 2 bits), 5bit (lowest 3 bits) and 4bit (lowest 4 bits).

This removes the subtle noise that appears when flat, illustration-style images are taken in with a scanner. However, gradations become step-like, so care is needed. Colors can also change, so if you do not understand the setting, we recommend leaving it at 8bit.

50%:50% net tile

A mesh pattern mixed at a 50% ratio is forcibly treated as a single color. Only very close colors are affected. This seemingly increases the number of colors.

Error diffusion

Error diffusion spreads the error caused by approximating a color to neighboring pixels, reducing the overall error of the image. Checking this option applies error diffusion (shown as "Error litter" in the English build's dialog). Its three sliders are as follows.

Slider	Description
Strength	The coefficient multiplied with the diffused error. It ranges from 0.000 to 0.500; at the maximum 0.500 there is no attenuation, and at 0.000 there is no diffusion. Around 0.430 generally gives good results. If sparse specks appear in the result, lowering this value slightly may improve it.
Cut-off threshold	Errors smaller than this value are forcibly treated as zero. This prevents subtle errors from being carried very far from where they occurred. A small value is usually sufficient. If discrete specks are noticeable, specifying a slight value may improve the result.
X:Y ratio	The ratio between the error diffused to the right (X direction) and downward (Y direction).

If error diffusion is not checked, these sliders have no effect.

Natural image algorithm (YJK algorithm)

Specifies the formula used for conversion for SCREEN10 and SCREEN12.

Option	Description
no use	Uses the most common method (the colors of the V9958 specification).
error fix (give brightness)	Tries every Y value and adopts the one with the least error.
MSX2+ POWERFUL BOOK ver.	Uses the formula of the ray-tracing program published in "MSX2+ Powerful Katsuyouhou" (a Japanese MSX book).
NTSC fix	Uses the NTSC YUV conversion formula with slightly adjusted parameters (R:G:B = 40:49:6).
error fix (give tone)	Tries every Y value and adopts the one with the least error. The criterion for the size of the error differs slightly from "error fix (give brightness)".

Recalculate J and K (JK recalculation)

In the SCREEN10 / SCREEN12 conversion, a provisional JK is calculated first and the Y values are derived from it. Checking this option makes BMP2MSX recalculate the JK values from the decided Y values.

5.4 Size Adjustment

Checking "Size Adjust" makes BMP2MSX adjust the size of the input image.

Resample

Specifies the sampling method used when enlarging or reducing.

Option	Description
normal	Adjusts by thinning out or stretching. Enlargement is jagged, and lines may break when reducing. Reduced images generally degrade severely.
antialiasing	Reduction uses the centroid of related pixels; enlargement uses bilinear interpolation. Enlargement looks blurred, reduction bleeds.

Size mode

Selects how the adjusted size (the aspect ratio decision) is determined.

Option	Description
output size forcibly	Ignores the dot ratio and outputs at the MSX dot width and height. No unnecessary gaps appear in the output.
look like aspect ratio on MSX	Adjusts the size so that the aspect ratio seen on Windows is reproduced when displayed on a real MSX. The image is fitted within the screen, so gaps may appear at the right or bottom.

Background color (Fill)

Specifies the color used to fill gaps in the output. Gaps occur when an image smaller than the MSX screen mode size is converted without size adjustment, and when adjusting to "look like aspect ratio on MSX". Note that this color also affects palette selection during color reduction.

5.5 Palette Settings

The "Palette settings" group at the bottom of the dialog covers the palette. In SCREEN8 and SCREEN12 output, which cannot use a palette, most items are unavailable.

Palette maximum selection (multi-distribution selection)

A check box that specifies how palette colors are decided. It has no meaning if there is no automatically decided palette at all.

- Unchecked: selects colors that are well scattered and heavily used, based on color usage
- Checked: simply selects colors in order of how much they are used. In this case, accent colors that appear only in small areas are hardly reproduced (colors of outlines and the like may not be selected)

Don't use color 0

Avoids using the color that becomes 0 in VRAM, replacing it with another color. Use this when the background should not become transparent with the VDP's TIMP (TPSET), or when the background color should not show through.

All custom palette (fixed palette)

Uses all palette colors as they are, without generating them automatically. Use this for game images and the like, where only fixed colors may be used.

The palette table and color editing

The palette table shows the palette contents as colors.

- Left click: selects that color
- Left drag: swaps the palette contents of the drag source and destination (the usage permission state is also swapped)
- Right click: changes the usage permission state

The selected palette number is shown as "COLOR = n", and its color can be changed with the RGB sliders.

Usage permission state

The usage permission state of the selected palette is specified in the list next to "COLOR = n".

Option	Description
automatic color	Lets BMP2MSX decide the color.
no use	This palette number is not used in the output file.
customize color	Uses the color specified with the RGB sliders as it is.

These states are shown as icons on the palette. The prohibited mark means "no use", and "HOLD" means "customize color".



Clicking "Set all pal" applies the usage permission state of the selected palette to all other palettes.

The whole palette can also be replaced at once with the buttons next to the table.

Button	Description
MSX1	Makes the palette MSX1-style colors.
MSX2 default	Makes the palette the MSX2 initial colors.

Sort

Sets whether palette values are sorted right after the palette is decided.

Option	Description
Ignore	Does not sort.
Incriment (auto only)	Sorts only palettes whose usage permission is "automatic color", in ascending order of luminance.
Decriment (auto only)	Sorts only palettes whose usage permission is "automatic color", in descending order of luminance.

Option	Description
Incriment	Sorts all palettes in ascending order of luminance.
Decriment	Sorts all palettes in descending order of luminance.

Palette output

Specifies where palette data is output. See Section 6.4 for the details of each destination.

Option	Description
no output	Does not output.
into the picture file (COLOR=RESTORE ver.)	Writes the palette at a fixed address in the BSAVE format, so that MSX-BASIC can load it with COLOR=RESTORE.
*.PL? (Graphsaurus ver.)	Outputs to a file in the PL? format used by BIT2's Graph Saurus.
*.CSV (CSV text file)	Outputs to a file in comma-separated text format.

Load palette / Save palette

The "Load Palette" button loads an output palette file (PL? format, or a BSAVE file containing a palette) and applies it to the palette table. The "Save Palette" button saves the current palette in PL? format. See Section 8.2 for details.

Forced zeroing

Forces a specific color of the original image to 0. Use it to specify the color to make transparent with the VDP's TIMP (TPSET). Note that when size adjustment is ON, contours blurred by the size adjustment may not be judged as the transparent color correctly.

Option	Description
no use	Does not perform forced zeroing.
appointed color	Forces the color specified with the zeroing color button (a full color) to zero.

Option	Description
No. -> No.0	
appointed (X,Y) color -> No.0	Forces the color at the specified position of the original image (after size adjustment) to zero. Enter the coordinates in the fields to the right of the zeroing color button, X on the left and Y on the right.

5.6 Default Settings and Reconvert Settings

The default settings dialog and the reconvert dialog look almost identical, but they differ as follows.

Aspect	Default settings	Reconvert settings
Purpose	The settings used for the first conversion after startup. Also used when files are dropped onto the BMP2MSX icon.	Shows the settings that were used to convert the displayed MSX display image.
Confirm button	Clicking "Default setting" adopts the contents as the default settings and closes the dialog.	Clicking "Reconvert" converts again immediately. If the output destination creates a preview, a new MSX display image is created while the current result is kept.
Persistence	Saved automatically even when BMP2MSX exits (BMP2MSX.B2M).	Lost when BMP2MSX exits.

Keeping your frequently used settings as the default settings saves you from setting them again at every startup.

If you like the reconvert settings, click "Copy to default" to copy them to the default settings. Checking "Always copy reconvert to default" in the default settings dialog copies them automatically at every reconversion.

In either dialog, "Save as" can save the settings to a B2M file, and "Load" can read one back.

Chapter 6 Screen Modes and Output Files

6.1 Screen Mode List

The screen modes BMP2MSX can output are as follows. They are selected under "MSX screen mode" in the settings dialog. The number in parentheses for the resolution is the number of lines when 192-line output is specified. With interlaced output, the vertical resolution is pseudo-doubled (512 lines for the 256-line modes).

Mode	Resolution	Colors	Notes
SCREEN2/4	256x192	2 colors per 8 dots, from a 16-color palette	Pattern + color table format. SCREEN4 shares the image format with SCREEN2.
SCREEN3	Equivalent to 256x192 (displayed as 64x48 blocks)	16-color palette	Multicolor mode.
SCREEN5	256x212 (192)	16-color palette	A 256-line mode exists.
SCREEN6	512x212 (192)	4-color palette	A 256-line mode exists.
SCREEN7	512x212 (192)	16-color palette	A 256-line mode exists.
SCREEN8	256x212 (192)	Fixed 256 colors	No palette. Colors are direct GRB332 values (green 3 bits, red 3 bits, blue 2 bits). A 256-line mode exists.
SCREEN8 EPAL(V9968)	256x212 (192)	256-color palette	For the V9968's extended palette mode (EPAL). Colors are chosen from a palette of RGB 5 bits each. A 256-line mode exists. See Chapter 7.

Mode	Resolution	Colors	Notes
SCREEN10/11	256x212 (192)	YJK system, 12499 colors	A 256-line mode exists. The 512-color palette part of SCREEN11 is not supported (see below).
SCREEN12	256x212 (192)	YJK system, 19268 colors	No palette. A 256-line mode exists.

A note on SCREEN10/11: SCREEN10/11 can use, in addition to YJK colors, 16 colors chosen from a 512-color palette. BMP2MSX does not support that palette part, and reproduces the image with the 12499 YJK colors only. In the settings dialog, SCREEN10/11 is treated as a single option.

A note on the 256-line modes: with the usual display method, only 212 lines can be shown. The programs generated by BAS output include a mechanism to scroll the screen with a key press so the whole image can be seen (Section 6.5). When using these modes in your own programs, combine them with the VDP's vertical scroll.

6.2 The BSAVE Format

The main conversion result is output in BSAVE format. A BSAVE file starts with a 7-byte header consisting of 0FEh and the start, end and execution addresses, followed by the data itself. From MSX-BASIC it can be loaded into VRAM with `BLOAD"filename",S`.

For modes that use a palette, when "Palette output" is set to "into the picture file (COLOR=RESTORE ver.)", the palette data is written at a fixed position in the file (that mode's palette table address). Executing `COLOR=RESTORE` after BLOAD in MSX-BASIC activates this palette. The palette table addresses are as follows.

Mode	Palette table address
SCREEN2/4	01B80h
SCREEN3	02020h
SCREEN5	07680h

Mode	Palette table address
SCREEN6	07680h
SCREEN7	0FA80h
SCREEN10/11	0FA80h

SCREEN8 and SCREEN12 have no palette, so no such writing takes place. Also, in the 256-line modes the palette is not combined into the image.

6.3 Output File Names

When converting with "Auto file name" checked, a file is created in the same folder as the source file, with the source file's extension replaced according to the following rules.

Condition	Extension form	Examples
Palette combined into the image	.SC?	TEST.SC5, TEST.SC7
Otherwise	.SR?	TEST.SR5, TEST.SR8
Interlaced	Two files, .S?0 and .S?1	TEST.S50, TEST.S51

The "?" is a character indicating the screen mode: 2 (SCREEN2/4), 3 (SCREEN3), 5, 6, 7, 8, A (SCREEN10/11), C (SCREEN12).

For the SCREEN8 EPAL modes, the extension is .SRP instead of following the rule above (.SP0 and .SP1 for interlaced output). See Chapter 7 for details of the EPAL modes.

For interlaced output, the file generated first is for page 0 (ending in 0) and the other is for page 1 (ending in 1). See Section 12.3 for how to display them.

If automatic file naming is not used, a save dialog appears for each conversion. For interlaced output it appears twice; specify a file name each time.

6.4 Palette Files

Palette data can also be output to a separate file from the image (the "Palette output" setting, Section 5.5).

Graph Saurus format (PL5 / PL7)

This is the format used by BIT2's Graph Saurus. It is a 256-byte file: 32 bytes holding 16 colors at 2 bytes per color (one byte packing red and blue, and one byte of green), repeated 8 times. When outputting, a dialog asks for the file name. By Graph Saurus convention, the file for SCREEN5 / SCREEN6 is called .PL5, and the one for SCREEN7 / SCREEN10 is called .PL7.

CSV format

A comma-separated text format. It starts with the header line ";MSX PALETTE DATA(&Hrb,&H0g)", followed by each color as a pair of hexadecimal bytes (the red/blue byte and the green byte), separated by commas. It covers 16 colors (4 colors for SCREEN6 only).

Loading palettes

With "Load Palette" in the settings dialog, you can load not only the PL? format files above, but also palettes from BSAVE files (*.S??) that contain a palette at a fixed address. For BSAVE files, only files whose palette is placed at that mode's palette table address (Section 6.2) can be read correctly.

6.5 BAS Output

When "BAS output" is checked in the settings dialog, an MSX-BASIC program with the same name as the BSAVE file but with the extension changed to .BAS is output at the same time. Loading and running this program in MSX-BASIC displays the converted result (Section 12.2).

The generated program starts at line number 100 in steps of 10, and looks roughly as follows (an example for SCREEN5 with the palette combined into the image).

```
100 DEFINT A-Z:COLOR 15,0,0:SCREEN 5
110 FOR I=0 TO 15:COLOR=(I,0,0,0):NEXT
120 BLOAD"TEST.SC5",S
130 COLOR=RESTORE
```

```
140 A$=INPUT$(1)
```

Depending on the mode and settings, lines like the following are added or changed.

- Interlaced: the SCREEN statement gains the interlace specification, SET PAGE and BLOAD appear in two pairs, and a line that cancels interlace at the end is added
- Palette not combined into the image: instead of COLOR=RESTORE, lines that pack the palette into a string and set it with COLOR= are included
- 256-line modes: a line that disables sprite display, and a line that scrolls the screen by manipulating VDP(24) with key input, are included
- SCREEN2/3: a machine-check line is included so the palette is only activated on MSX2 and later
- SCREEN8 EPAL: lines that enable EPAL, and lines and DATA statements that set the palette (256 colors), are included (Chapter 7)

6.6 Saving a Preview as BMP

Choosing Preview → Save BMP saves the displayed preview (the MSX display image) as a BMP file. Use this when you want to check or share the MSX-format conversion result on the Windows side.

Chapter 7 Output for the V9968

7.1 What Is the EPAL Mode

The V9968 is a VDP designed as an upward-compatible successor to the V9958, and is used in the MSX2++ and the V9968 cartridge, among others. Its Graphic7 mode (equivalent to SCREEN 8 in MSX-BASIC) has an extended palette mode (EPAL). With EPAL enabled, each byte in VRAM is interpreted not as a direct GRB332 color, but as a number into a palette of 256 colors. Each palette color is specified with 5 bits each for R, G and B (32,768 colors in total), so smoother gradations can be expressed than with SCREEN8's direct colors.

BMP2MSX supports output for this EPAL mode. The "MSX screen mode" list in the settings dialog has the following two entries in addition to the existing modes.

Mode	Resolution	Notes
SCREEN8 EPAL(V9968)	256x212 (192)	The number in parentheses is the number of lines for 192-line output
SCREEN8 EPAL(V9968) (256lines)	256x256	The 256-line version

The EPAL modes are specific to the V9968 and cannot be displayed correctly on an MSX with a conventional VDP (V9938 / V9958). If the display target is a conventional MSX, convert with an existing mode such as SCREEN8 (Chapter 6).

7.2 How Conversion Works

For an EPAL conversion, the colors used in the source image are first counted in a color space of 5 bits each for R, G and B, and the palette of colors to use is decided automatically.

- If the image uses at most 256 colors: those colors are adopted into the palette as they are, in order of decreasing frequency of use
- If it uses more than 256 colors: the colors are reduced to 256 by the median-cut method (groups

of colors are split repeatedly, and a representative color of each group becomes a palette color)

The palette is always generated automatically; a fixed palette cannot be specified. Each dot is assigned the palette color nearest under the same color-difference measure as the other palette modes. The dither pattern and error diffusion settings of the settings dialog are also effective at this time (Section 5.3). Tile conversion cannot be used with the EPAL modes.

The palette settings concerning number 0 — "do not use number 0" and forced zeroing — are also effective for the EPAL modes. With "do not use number 0", palette number 0 is not used, and colors subject to forced zeroing are assigned to number 0 (Section 5.5).

7.3 Output Files

The image data of the conversion result is output in BSAVE format, in the same shape as SCREEN8. The contents are not direct colors, however, but palette numbers of one byte per dot. With automatic file naming, the extension is .SRP, or two files .SP0 and .SP1 for interlaced output (Section 6.3).

The palette itself is not contained in the image file. To pass the palette to the V9968, use BAS output. **When converting in an EPAL mode, be sure to check "BAS output" in the settings dialog.** The generated BAS program contains roughly the following steps.

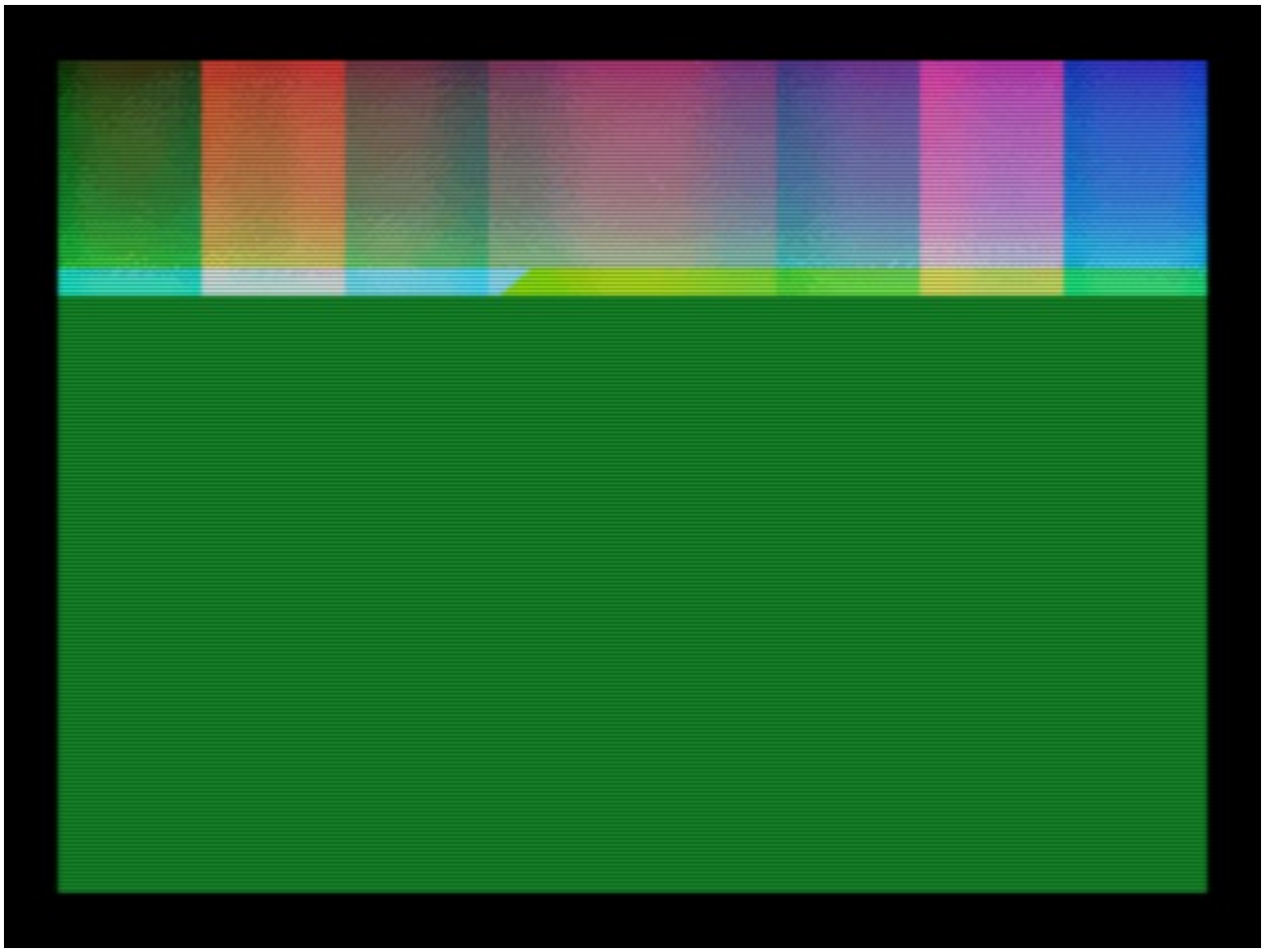
- Initialize the screen with SCREEN 8
- Write 16 to VDP register R#20, enabling EPAL
- Reset the palette number to 0, then read the R, G and B values (each 0 to 31) for the 256 colors from the DATA statements at line numbers 60000 and onward, and write them to the palette registers in order
- Load the image file (.SRP) with `BLOAD"filename",S`

To display the result, it is easiest to load and run this BAS program in MSX-BASIC (Section 12.2). To use the result in your own program, enable EPAL and set the palette, then load the image file with BLOAD (Section 12.1).

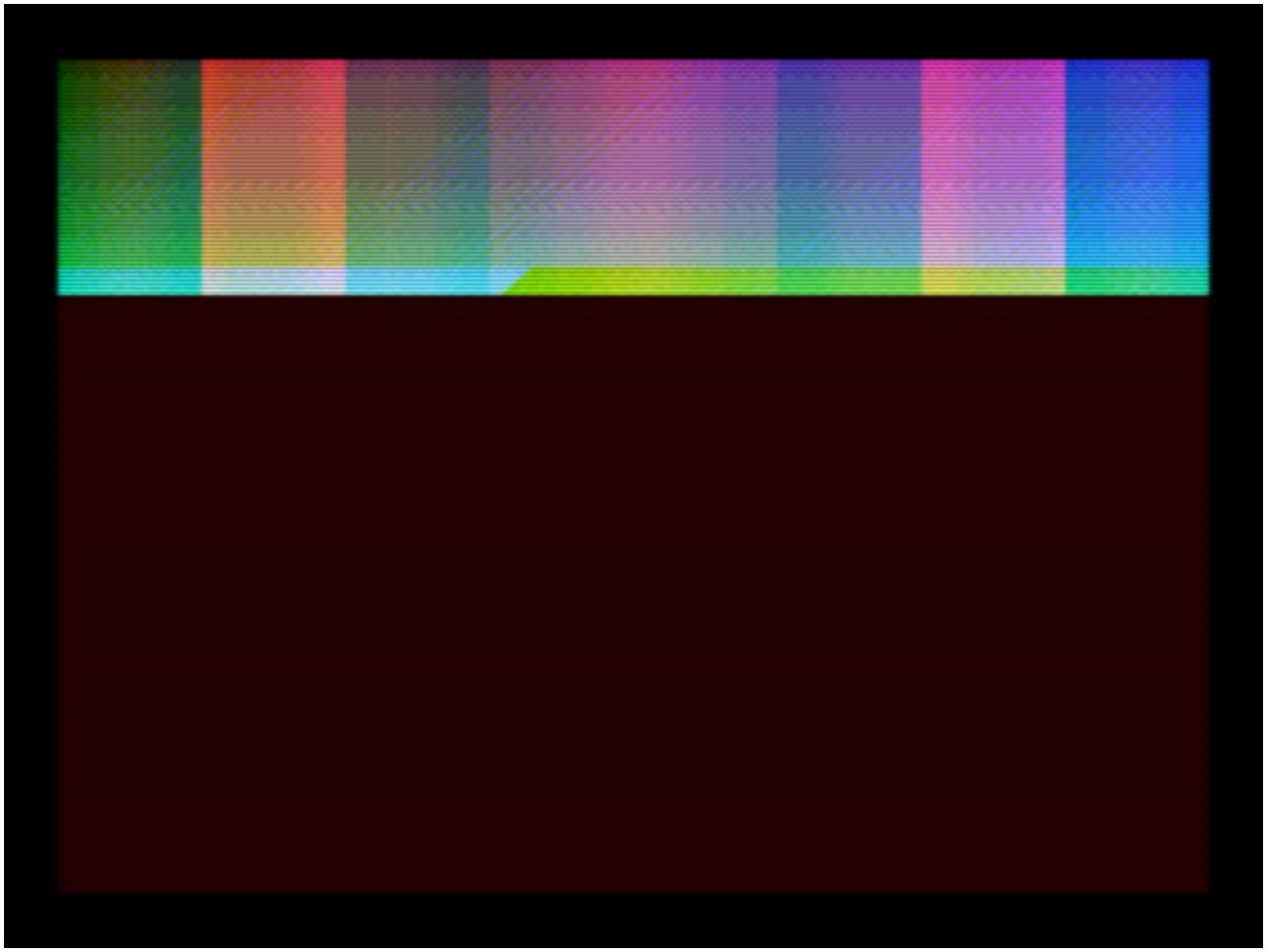
Note that the storage area for the EPAL palette was appended at the end of the settings file structure, so settings files saved by older versions can be loaded as they are.

7.4 Display Examples

The following are examples of the same photographic image converted in the EPAL mode and in conventional SCREEN8, displayed on openMSX emulating a V9968.



In the EPAL version, gradations are displayed smoothly, with 5 bits of gradation for each of R, G and B.



In the conventional SCREEN8 version, steps in the gradations and graininess stand out. The more colors an image contains, the greater the effect of the EPAL mode.

Chapter 8 Palette Editing and Color Calibration

8.1 Editing the Palette

In modes that use a palette (SCREEN2/4, 3, 5, 6, 7, 10/11), the 16-color palette used for conversion (4 colors in SCREEN6) can be edited manually in the palette settings of the settings dialog. The basic operations are as described in Section 5.5.

- Click the palette table to select a color, and adjust it with the RGB sliders. Set that palette's usage permission state to "customize color"
- A number whose usage permission state is "no use" will not be used in conversion
- Checking "All custom palette" disables automatic selection entirely and uses only the colors set in the table
- Dragging swaps colors, and right-clicking changes the usage permission state

To find out which colors are used in the original image, the eyedropper of the original image window (Chapter 11) is useful. Decide palette values while looking at the MSX palette reference values of the colors picked with the eyedropper.

8.2 Loading and Saving Palettes

The palette settings of the settings dialog have buttons for exchanging palettes with files.

Button	Description
Load Palette	Shows a dialog for opening a palette file (*.PL?) or a BSAVE file containing a palette (*.S??), and applies the loaded palette to the palette table. For BSAVE files, only files whose palette is placed at that mode's palette table address (Section 6.2) can be read correctly.
Save Palette	Saves the current palette table to a file in Graph Saurus format (PL? format, Section 6.4).

For outputting the palette to a separate file at conversion time (the Graph Saurus format / CSV format choices of "Palette output"), see Sections 5.5 and 6.4.

8.3 Color Calibration

Choosing Setting → Color calibration opens the color calibration dialog.

BMP2MSX has tables that convert MSX gradations (each palette color component 0-7, each SCREEN8 component, each SCREEN12 color 0-31) to Windows gradations of 0-255, and uses them for preview display and conversion. However, the actual appearance differs between a real MSX and Windows. Color calibration lets you adjust these conversion tables to match the appearance of the real machine.

There are three kinds of targets, switched with SCREEN MODE at the top of the dialog.

Target	Contents
RGB 512 Colors palette	The conversion table for palette colors (R, G, B each 0-7).
SCREEN8	The conversion table for SCREEN8 direct colors (red and green 0-7, blue 0-3).
SCREEN12	The conversion table for SCREEN12 (YJK, R, G, B each 0-31).

Selecting a target displays a color bar of its gradations. At the bottom of the dialog, select an element (such as RED 0 to 7) and adjust the Windows-side value (0-255) for that gradation with the slider. "Undo" cancels the last change, and "Load default" returns everything to the initial values. Confirming with OK saves the adjustments to BMP2MSX_CUSTOM.BIN.

For adjustment, it is effective to display the gradation chart on a real MSX using the accompanying tools (the PAL512 set in tool/Calibration) and compare it with the Windows preview while fine-tuning.

Chapter 9 Scripts

9.1 The Script Window

BMP2MSX has a script feature. By describing fixed conversion procedures in text, conversion work can be semi-automated. For example:

- Using SCREEN8 images for MSX2 and SCREEN10 images for MSX2+ and later (doing the same conversion twice by hand, reloading settings files, is tedious)
- Converting SCREEN5 images with various settings to compare the results

Choosing File → Script mode opens the script window. Its menus are as follows.

Menu	Item	Description
File	New	Creates new script text.
File	Open	Loads a script file (.TXT).
File	Save as	Saves the script text to a file.
File	Exit	Closes the script window.
Run	—	Runs the written script.

Load or type the script text and click "Run" to execute the script. If you drop an image file onto the script window and run the script, that file name is automatically set as the conversion target.

9.2 Script Commands

A script is written as a list of commands. The available commands are as follows. Each command has an English name and a Japanese name, and either may be used. English command names are case-insensitive.

Command (English / Japanese)	Argument	Meaning
LOADSETTING / 設定読込	"file name"	Loads the B2M file specified by the file name. Include the extension. The file name must be enclosed in half-width " (double quotes).
CONVERT / 変換処理	None	Converts the file set as the target, using the current settings. When the script is run by dropping a file, that file name is set automatically. When run with "Run", a file name must be specified with SETFILENAME, or an error occurs.
SETFILENAME / ファイル名	"file name"	Specifies the file name to convert.
; (comment)	To end of line	Everything after ; to the end of the line is ignored. Used for comments. ; is a half-width semicolon; full-width characters cannot be used.
SCREEN / 画面モード	Output mode number	Specifies the output mode number. The numbers 2/3/4/5/6/7/8/10/11/12 can be specified, the same as the first argument of the MSX-BASIC SCREEN statement. Specifying 4 gives a conversion for SCREEN2, and 11 gives one for SCREEN10.
MSG / メッセージ	"message"	Shows the message in a small window. The message must be enclosed in half-width ", and cannot itself contain ".
SETFLAG / フラグ	Flag name=value	Sets the setting indicated by the flag name (Section 9.3).
SETTING / 設定画面表示	None	Shows the conversion settings dialog (the settings dialog).
PALETTE / パレット	Palette number=red,green,blue	Sets the color of the specified palette number of the MSX palette.
PALMODE / パレッ	Palette	Sets the usage permission state of a palette. For the value, use the

Command (English / Japanese)	Argument	Meaning
ト動作	number=value	constants of Section 9.3.
CUTERR / 切り捨て誤差	Value	Specifies the cut-off error (cut-off threshold).
GOSAVAL / 誤差拡散係数	Value	Specifies the error coefficient (strength). The value is an integer from 0 to 500, and 1/1000 of it is the error coefficient.
SELCOLMODE / 選色モード	Value	Specifies the palette decision algorithm (multi-distribution or scatter).
DITHERADD / ディザ加算	Value	Specifies the dither adding mode.
ALGOMODE / 自然画生成	Value	Specifies the natural image generation algorithm.

Note that the SCREEN command accepts only numeric mode numbers, so the V9968 EPAL modes (Chapter 7) cannot be selected from this script feature. To use the EPAL modes from a script, use the command-line version (Chapter 10), which adds SCREEN 8P / SCREEN 8P256 as an extension.

An example script follows. It converts with a prepared settings file for SCREEN10, and then also for SCREEN8.

```
; Make both SCREEN10 and SCREEN8 versions
LOADSETTING "photo_sc10.B2M"
SETFILENAME "photo.bmp"
CONVERT
SCREEN 8
CONVERT
MSG "done"
```

9.3 Flags and Constants

The following values can be specified as the flag name of the SETFLAG command. Each flag has an English name and a Japanese name.

Flag (Japanese / English)	Meaning
誤差拡散 / GOSA	Whether to perform error diffusion.
インターレース / INTERACE	Whether to perform interlaced output (the English name is spelled INTERACE).
固定パレット / PALNOCHG	Whether to use all palettes as they are.
サイズ調節 / RESIZE	Whether to scale the size to the MSX resolution.
自動ファイル名 / AUTONAME	Whether to generate the file name automatically from the source file name.
色差再計算 / COLORRECALC	Whether to recalculate the JK values in natural image mode.
ゼロ使用不可 / ZERONOUSE	Whether color number 0 must not be used.
タイル / TILE	Whether to use 50%:50% ratio tiles as single colors.

In addition to numbers, the following constants can be specified as the value of each command.

Constant	Related command	Meaning
TRUE / ON	SETFLAG	Enabled.
FALSE / OFF	SETFLAG	Disabled.
NOUSE / 未使用	PALMODE	This palette is not used.
AUTO / 自動	PALMODE	The color of this palette is decided automatically.
CUSTOM / 指定	PALMODE	This palette is used with the specified color.
SCATTER / 分散	SELCOLMODE	Selects well-scattered colors, based on the most-used color.

Constant	Related command	Meaning
MAX / 多分布	SELCOLMODE	Prefers the most-used colors.
ROTATE / 回転	DITHERADD	Rotate.
NONE / 同位相	DITHERADD	Same phase.
X / X ずらし	DITHERADD	Shift in X.
Y / Y ずらし	DITHERADD	Shift in Y.
NONE / 無し	ALGOMODE	None.
RECALC / 再計算	ALGOMODE	Error correction (brightness priority).
POWERFUL / パワフル	ALGOMODE	The Powerful Katsuyouhou ray-tracing formula.
NTSC2 / NTSC 改	ALGOMODE	The modified NTSC YUV-to-RGB conversion parameters.
COLOR / 色合い	ALGOMODE	Error correction (tone priority).

9.4 Numbers and Expressions

Parameters that specify numbers, such as the error coefficient, can also be given as formulas. Only expressions that can be evaluated as constants are allowed; variables and the like cannot be used.

Besides decimal numbers, binary numbers with a trailing B (for example, 101011B) and hexadecimal numbers with a trailing H (for example, 0FFH) can be written.

The operators that can be used are ( ), *, +, /, %, &, ^, ~, |, AND, OR, XOR and NOT. Their meanings are the same as in C and MSX-BASIC.

Chapter 10 The Command-Line Version

10.1 What Is the Command-Line Version

BMP2MSX has a command-line version, bmp2msx-cli, that runs on Linux. Its conversion processing uses the same core as the Windows version, so with the same settings and the same image, output files byte-identical to the Windows version's are obtained. The GUI is not ported, so features that involve screen interaction, such as previews, are not available (Section 10.7).

The files of the command-line version are placed in the linux folder of the distribution repository.

10.2 Building

Only g++ (C++17) is required to build. Running the following in the linux folder produces the executable bmp2msx-cli.

```
cd linux
make -j
```

Intermediate files are placed in the obj folder, and incremental and parallel builds are supported. make clean removes the intermediate files and the executable. This build does not affect the Windows version.

10.3 Basic Usage

To convert a single image, specify options and the image file name.

```
./bmp2msx-cli [options] photo.bmp
```

To run a script, use --script (Section 10.5).

```
./bmp2msx-cli --script convert.txt --input photo.bmp
```

--input is the image file passed as the conversion target when running a script. The exit status is 0 for success, 1 for usage or startup errors, and 2 for conversion or script errors.

Settings start from the application's default settings. The settings file (BMP2MSX.B2M) is not loaded

automatically. To load one, specify it with --load-setting; options given on the command line then take precedence. If BMP2MSX_CUSTOM.BIN (the color calibration data) is present in the current folder, it is loaded at startup, as in the Windows version.

10.4 Options

The options cover almost every conversion-related item of the Windows version's settings dialog. The main ones are as follows. All options can be checked with --help.

Option	Description
--screen	The screen mode. 2/3/4/5/6/7/8/10/11/12 can be specified; append -256 for the 256-line versions (for example, 8-256). The EPAL modes are 8p / 8p-256.
--palette-output	Where the palette is output (bsave / pl / csv / none).
--basic	Enables BAS output.
--interlace	Performs interlaced output.
--lines-192	Performs 192-line output.
--load-setting	Loads a settings file (.B2M).
--output-dir	Specifies the output folder.
--script	Runs a script file.
--input	The conversion target file for script execution.

In addition, the resize and resampling methods, the error diffusion coefficients, the dither pattern, the color selection algorithm, individual palette color specifications, tiles, forced zeroing and more can also be specified with options.

10.5 Running Scripts

The script language is the same as the Windows version's script feature (Chapter 9), and the English and Japanese command names and expression evaluation work as they are. The differences from the Windows version are as follows.

- As an extension of the SCREEN command, SCREEN 8P / SCREEN 8P256 can be used to specify the EPAL modes (the Windows version's scripts cannot specify the EPAL modes)
- The SETTING command does not open a dialog; it prints the current settings to standard output
- The MSG command prints the message to standard output
- File names are always determined automatically. AUTONAME is accepted, but no file name is ever asked for

Scripts and file names are in UTF-8. A script file that cannot be interpreted as UTF-8 is assumed to be Shift_JIS and converted automatically, so existing scripts created on Windows can be run as they are.

10.6 Output File Names

Output file names are derived from the source file name by the same rules as the Windows version's automatic file naming (Section 6.3, Chapter 7), and the files are written to the same folder as the source file. With --output-dir, they are written to that folder instead.

When the palette output is PL or CSV, the Windows version asks for a file name when saving, while the command-line version determines it automatically from the source file name. PL files receive the extension for their mode family (.PL5 for the SCREEN5/6 family, .PL7 for the SCREEN7/10 family, .PL2 / .PL3 for the SCREEN2/3 family), and CSV files receive .CSV.

10.7 Notes and Limitations

The following features are not included in the command-line version. All of them are GUI-only features.

- Preview display, and saving previews as BMP
- Drag & drop, and input from the clipboard
- The settings, color calibration and eyedropper dialogs (the color calibration data file, BMP2MSX_CUSTOM.BIN, itself is still loaded)

For image loading, BMP files are loaded with the same processing as the Windows version. PNG and JPEG are loaded using stb_image. PNG is lossless, so it gives the same results as BMP. JPEG decompression results differ slightly between decoders, however, so results may differ from the Windows version (WIC) by rounding in a few dots.

Chapter 11 The Original Image Window

11.1 The Original Image Window

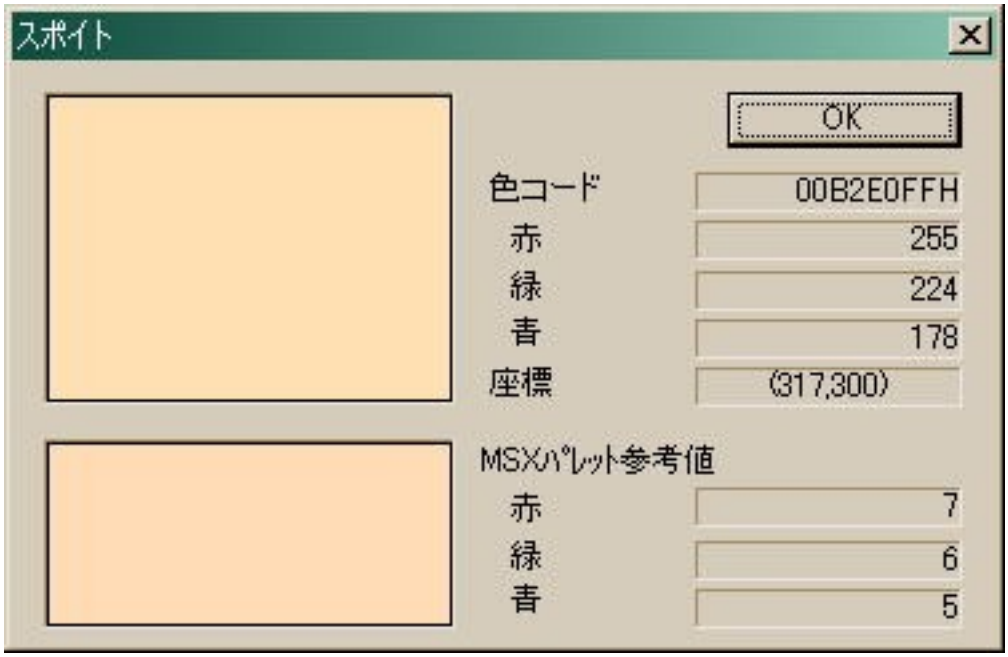
After converting an image, while an MSX display image (preview) is shown, choosing Preview → View → Show original (Ctrl+O) displays the complete original image, before size adjustment, in a separate window (the original image window). This differs from the "source" display switched with the Space key in the main window (Section 4.4), which shows the image right after size adjustment; this window shows the image exactly as loaded.



This window's menus are "Exit" and "Function"; the Function menu selects the x1 view or x2 view. The window itself can also be resized freely.

11.2 The Eyedropper

Moving the cursor over the original image window changes the cursor to an eyedropper shape. Clicking the image in this state opens a dialog showing the color information of the clicked position.



The information shown is as follows.

Item	Contents
Upper-left color	The color exactly as picked by the eyedropper.
Lower-left color	That color approximated to the MSX 512-color palette.
Color code	The value representing the color, shown as 8 hexadecimal digits followed by H.
Red / Green / Blue	The value of each component, in decimal, each in the range 0-255.
Coordinates	The pixel position, with the top left of the image as (0,0), X to the right and Y downward as positive.
MSX palette reference values	The red, green and blue values of the approximated color (each 0-7).

In conversions that use a palette, the palette decision greatly affects the result. If you do not want to leave it to automatic selection, decide the palette values in the settings dialog while referring to the color information obtained with the eyedropper (Chapter 8).

While the eyedropper dialog is open, the original image window cannot be operated, but the main window can. You can also open the settings dialog and work with it side by side with the eyedropper

dialog.

Chapter 12 Displaying the Result on an MSX

To display a converted result (a BSAVE-format file) on an MSX, you can either use BLOAD from MSX-BASIC, or use the program generated by BMP2MSX's BAS output. The accompanying Load.txt also contains sample programs; please refer to it as well.

12.1 Displaying with BLOAD

SCREEN5 (no interlace, palette combined into the image)

```
10 SCREEN 5
20 COLOR 15,0,0
30 BLOAD "TEST.SC5",S
40 COLOR=RESTORE
50 GOTO 50
```

Line 30 loads the image, and COLOR=RESTORE at line 40 activates the palette loaded with the image.

SCREEN8 (no interlace)

```
10 SCREEN 8
20 BLOAD "TEST.SR8",S
30 GOTO 30
```

SCREEN8 has no palette, so COLOR=RESTORE is not needed.

SCREEN12 (no interlace, MSX2+ and later)

```
10 SCREEN 12
20 BLOAD "TEST.SRC",S
30 GOTO 30
```

SCREEN12 has no palette either.

12.2 Using BAS Output

If you convert with "BAS output" checked in the settings dialog, a display program (.BAS) is output at the same time (Section 6.5). In that case, simply loading and running that .BAS file in MSX-BASIC displays the result. This is convenient when the display procedure is complicated, as with interlace or the 256-line modes.

12.3 Displaying Interlaced Images

Interlaced output generates two files, one for page 0 and one for page 1 (for example, TEST.S70 and TEST.S71). To display them, specify interlace mode with the SCREEN statement and load each file into its page.

```
10 SCREEN 7,,,,,3
20 COLOR 15,0,0
30 SET PAGE 0,0
40 BLOAD "TEST.S70",S
50 SET PAGE 1,1
60 BLOAD "TEST.S71",S
70 COLOR=RESTORE
80 GOTO 80
```

The last argument of line 10 specifies the interlace mode. The palette combined into the image loaded later (the page 1 side) is activated by COLOR=RESTORE. If you use BAS output, this procedure is also generated automatically.

Alternatively, graphics viewers that can load the BSAVE format can be used. Among the tools that were sold commercially, Graph Saurus and DD Club can load them. Models that came with graphics tools may be able to load them with those tools. However, only images of the screen modes those tools support can be loaded correctly. For interlaced images and SCREEN2/3 images, writing a program is the reliable way.

Chapter 13 Distribution and Support

13.1 Copyright and Distribution

Copyright

BMP2MSX is freeware. Its copyright is held by HRA!.

Distribution

Distribution is permitted when all of the following conditions are met.

- No money is involved in the distribution. However, the actual cost of distribution media and the like is excluded, as are cases where permission has been obtained from HRA!
- The contents of the distribution archive are not modified
- The contents of the disclaimer are conveyed to the recipients of the distribution
- When publishing in a homepage, magazine or the like, HRA! is contacted in advance about the publication

Disclaimer

Those who cannot agree to the following conditions may not use BMP2MSX.

- Even if the user suffers any damage from the use of BMP2MSX, HRA! bears no responsibility whatsoever
- Even if BMP2MSX has defects, HRA! is under no obligation to fix them

13.2 Support and the Latest Version

As stated in the disclaimer, there is no obligation to fix bugs, but the intention is to fix and improve as much as possible. If you notice any bugs or have any requests, please let us know.

The latest version and the source code are published on the bmp2msx repository on GitHub.

- GitHub: https://github.com/hra1129/bmp2msx

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BMP2MSX USER'S MANUAL

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